

Sreenidhi Institute of Science and Technology

Department of Computer Science and Engineering

B. Tech CSE III Year I Sem Section – D

Computer Networks Assignment

Date of submission: 15/11/2025

- 1) Explain about Network Topologies with neat diagram.
- 2) Illustrate the functionalities of each layer of ISO/OSI model and also draw the model with neat diagram.
- 3) Illustrate the functionality of various layers in TCP-IP and also draw the model with neat diagram.
- 4) Categorize various Networks and explain each network.
- 5) Classify Transmission media and explain each media with neat diagram
- 6) Assume that $g(x)$ is CRC-4, which equals $X^4 + X + 1$, and the source data M is 10110011. Use CRC algorithm and find out CRC checksum
- 7) Construct the even parity Hamming code word for a data byte 1001101
- 8) Elaborate “Go Back n” and “Selective Repeat” Sliding Window Protocols.
- 9) Draw HDLC and PPP frame format and explain each field.
- 10) Differentiate between pure ALOHA and slotted ALOHA.
- 11) Illustrate CSMA / CD protocol with flow chat and also explain CSMA protocols.
- 12) Explain about MAC address, IEEE 802.3 and IEEE 802.11.
- 13) List different types of Bridges, write short notes on each Bridge.
- 14) Differentiate between Virtual Circuit and Datagram subnets.
- 15) Describe shortest path routing algorithm with example.
- 16) Describe Distance Vector routing algorithm with example.
- 17) What do you understand by Tunneling?
- 18) Draw IPV4 header format and explain each field
- 19) Let us consider IP address 195.16.100.130/26, find out Network address, host addresses, Broadcast address and subnet mask
- 20) Write short notes on ICM, ARP, RARP and DHCP.
- 21) Describe congestion control techniques in Datagram subnet.
- 22) Explain about principles of congestion control.
- 23) Describe leaky bucket and Token Bucket Algorithm
- 24) Write short notes on connection establishment and connection release.
- 25) Draw TCP & UDP header format and explain each.
- 26) Write short notes on DNS, FTP, SMTP, HTTP and www.